

Matthieu Ortscheit

Thesis

Title project: The effect of a ketogenic diet on sleep, mood and cognition in healthy adults: a pilot study

The ketogenic diet (KD), a low-carbohydrate and high-fat diet, has seen a growth in interest, within and outside of academia, in recent years. Existing research has safely introduced the KD diet to various populations where various positive physiological and cognitive effects of the KD on populations suffering from illnesses (obesity, epilepsy, and cognitive impairments) were found. Research on the effects of the KD on healthy adults has shown mixed results. This study aimed to determine the effects of the KD on sleep, mood, and cognition in healthy, normal-weight adults. Normal-weight female participants completed this non-randomized pilot study where they followed the KD for 4-week period. Sleep and mood were assessed using self-report measures, while cognitive function was assessed using a computer-based task. The dependent variables were assessed at baseline and post-intervention. We hypothesized that participants would show differences in sleep quality, mood, and cognition following a KD for four weeks.

Leiden University

External supervisor: Dr. Eline Dekeyster

UU Supervisor: Dr. Maarten van der Smagt

Internship Abstract
Matthieu Ortscheit
Matching Content Presentation to the User: A Guide

Ancora Health, a digital healthcare start-up, aims to enhance its programs by tailoring content delivery to match users' knowledge, learning ability and preferences, thus boosting engagement and effectiveness. This project under the Health Engagement Team focused on this customization in mobile health, considering various content formats such as written media, visual media, auditory media, quizzes and chatbot technology. Recognizing that user characteristics differ based on factors like age, gender, culture, preferences, ability and knowledge, the project produced a research report and power point presentation which provided guidelines for tailoring content presentation to match the user. Based on thorough literature research, the theoretical report summarized the research on mobile health, provided a competitor analysis, and guidelines on how to best present various forms of content depending on user characteristics. Ultimately, this project had the goal of delivering actionable insights for future data-driven program enhancements.

Name company: Ancora Health
External supervisor: Rahul Gannamani
UU Supervisor: Suzanne Hut

Name Student: Lucie Pearl Russell Kearns

Internship or Thesis: Internship

Title project: Cognitive Training for the Elderly: How Can the Literature Inform Updates to the SilverFit 3D Exergames?

Cognitive training (CT) is a type of intervention method designed to optimize cognitive function through structured practice on tasks relevant to various areas of cognitive functioning. The current report investigates the literature on the use of these interventions to improve cognitive outcomes in elderly populations with the goal of providing a collection of evidence-based literature that can be used to inform updates to a SilverFit exergame device. Non-systematic literature searches were conducted and studies that involved healthy adults with an average age of 70 or above, or included older adults with psychiatric and neurological health conditions were selected. In total, 11 studies explicitly testing task- or game-based “cognitive training” methods were selected for comparisons to the SilverFit games. The evidence gathered suggests that while the device’s current cognition menu likely has a sufficient and diverse selection of memory games, various other cognitive processes may be underrepresented. Recommendations for additional games and the structure of a cognitive training program are provided.

Name of company or University: SilverFit

External supervisor: Ame Harris

Internal supervisor: Larissa Meijer

Name: Lucie Pearl Russell Kearns

Internship or Thesis: Thesis

Title project: Me, Myself, and AI: Investigating the Influence of Self-Esteem on Adoption of Human vs AI Suggestions

With the increasing use of Artificial Intelligence (AI) in many domains, collaborations between humans and algorithms are becoming more common in decision-making scenarios. Several factors impact if an individual chooses to accept or reject the suggestions of AI agents. However, many of these variables are context dependent. In human-human relationships, evidence suggests that an individual's level of self-esteem can predict their likelihood to accept the advice of others. As global self-esteem is considered to be stable across situations, this study sought to replicate this effect with both human and artificial teammates and investigate if the effect is stronger with a human teammate than an AI one. The bin-packing game was utilized as the experimental task. Contrary to our hypotheses, regression analyses revealed no significant relationship between self-esteem, as measured with the RSES and the Problem-Solving Confidence subscale from the PSI, and likelihood to adopt teammate solutions. Exploratory analyses weakly suggest that these variables may have demonstrated a positive relationship solely in the AI teammate group. These results contradict previous findings regarding advice-taking behaviour among humans and bring into question the stability of this relationship.

Name company or University: Utrecht University

UU Supervisor: Talha Ozudogru

Name Student: Xuanwei Luo

Internship or Thesis: Thesis

Title project: . Correlation analysis of Stress-related HRV in dyads of caregiver and child with ASD

(Abstract, 100-150 words)

Name company or University: TU Eindhoven

(If applicable) External supervisor: Jing Li

UU Supervisor: Krista Overvliet

The phenomenon of stress synchronization between children with Autism Spectrum Disorder (ASD) and their caregivers during interactive tasks presents a compelling field of investigation, notably within the context of using non-invasive measures such as Heart Rate Variability (HRV) for assessment. This study aims to explore the dynamics of stress synchronization, leveraging HRV as a physiological marker to objectively evaluate stress levels in high-functioning ASD children aged 6-12 and their primary caregivers. Employing a structured experimental design, this research uses interactive Lego building tasks of varying difficulty to induce stress, alongside relaxation phases with calming music and breathing exercises, to create a baseline.

The interaction sessions, which are meticulously recorded and examined for HRV data, aim to discover patterns of stress synchronization within ASD family dynamics. Preliminary findings suggest a complex interaction between task-induced stress levels, emphasizing the need for a careful balance between challenge and support in order to promote positive developmental outcomes in children with ASD. This research will not only contribute to the understanding of stress and emotional synchronization in families with ASD but also highlights the potential of HRV as an effective real-time tool for measuring emotional and physiological responses, opening the door to novel therapeutic approaches.

Name Student: Elise Manizer Likhatcheva

Internship or Thesis: Thesis

Title project: Implicit Attitudes and Comfort with Social Touch. A quantitative approach using the implicit association test

This study, which investigates the relationship between comfort with social touch and implicit attitudes towards touch, was conducted using a robust methodology. We employed the Implicit Association Test (IAT), a tool that measures automatic associations, to uncover subconscious attitudes that traditional self-report methods might not detect. Participants (N=66) were asked to complete a Likert scale questionnaire to assess their comfort levels with social touch and then take the IAT. Our analysis revealed that comfort levels accounted for only 8% of the variance in D-scores, indicating a weak but significant negative correlation. This suggests that individuals with higher comfort levels do not necessarily hold more positive implicit attitudes towards social touch. These findings underscore the complexity of social touch perception and highlight the importance of using implicit measures to understand attitudes towards social touch without the biases inherent in self-report methods. Future research should consider additional factors such as cultural background and past experiences to explain these relationships further.

University: Utrecht University

UU Supervisor: Anouk Keizer

Name Student: Qianze Zhang

Internship or Thesis: Internship

Title project: Hand Tracking and Analysis in Dyadic Task Set-up

Abstract: This report presents a review of the scientific theoretical background pertaining to the synchronization behavior under time-pressure, along with a series of pilot experiments focusing on hand detection. This internship combined with a master thesis examines the significance of eye and body movements in human synchronization behavior and investigates gaze and the proximity of hand positions through experiments. We extended the experimental set-up outlined by Hessel et al. (2023) and designed an experiment for recording eye movements and tracking hand positions under time pressure. The objective of this report is to validate the methodology for hand tracking and establish a theoretical framework in anticipation of forthcoming formal experiments.

Name company or University: Utrecht University

UU Supervisor: Dr. Roy Hessels

Name Student: Qianze Zhang

Internship or Thesis: Thesis

Title project: How does synchronous behavior appear under time pressure during dyadic collaboration.

Abstract:

Synchronous behavior is a behavior that matches in time. People often get closer (both physically and visually) to better complete collaborative projects. This project designed a task in which a pair of people worked together to copy a Lego Duplo model, changing only whether it was completed within a time limit. Wearable eye trackers and gloves with Aruco markers were used to track the participants' gaze and hand positions. The results showed that their hand positions decreased slightly, but the distance between their gaze points did not change due to time pressure, nor did it decrease for better collaborative performance. In addition, their joint gaze was related to the task goal and their performance in the untimed trials improved significantly, but this improvement may be the result of implicit cooperation. The results suggest that the subjects may not have felt enough time pressure during the experiment, and that the indicators of synchronous behavior should not only be proximity but should include more complex behavioral analysis.

Name company or University: Utrecht University

UU Supervisor: Dr. Roy Hessels

Minahil Azhar - 1463225

Internship: 12.5 ECTS

Title: Designing the Interface for CureDoc

I am doing my internship at CureMD a US based software company working in digital health. They provide multiple innovative solutions to different stakeholders in the field of health care. During my internship I was part of a new project under the name of CureDoc. Which is very similar to Chat GPT an AI assistive machine but only provides medical and health advice. The main goal was to design an interface for CureDoc to do so I did a detailed comparative analysis with Chat GPT, Monica, Gemini, and Perplexity. I looked at their layout, use of Icons, Colors, Landing page, navigation bar, profile and other interface elements that could help us in the design process for CureDoc. The next steps included creating prototypes and then finalizing the design which is to be done by the approval of CEO and UX/UI team leads. Emphasis is placed on developing a user-friendly, efficient, and aesthetically appealing interface that reduces cognitive load and maintains consistency.

Company Name: CureMD

External Supervisor: Ken Edwards

UU Supervisor: Dr. Chris Janssen

Name Student: Imara Hesselink

Internship or Thesis: Thesis (27.5 ECTS)

Title project: Picking Brains While Packing Bins: Experts Rely on Both Humans & Artificial Intelligence When Solving the Bin-Packing Problem

Abstract:

The present research explored how domain expertise affects people's preference for input from a human or AI, and if expertise impacts people's reaction to an AI's and a human's errors. To do so, we conducted an experiment where experts and novices in problem-solving tackled the bin-packing problem. A human or AI teammate gave participants solutions to this problem, with some solutions being erroneous. Results showed that experts accepted a teammate's inputs more than novices, regardless of who the teammate was (human vs AI), or if they had made errors. This indicates that expertise increases people's acceptance of a teammate's input, and that it does not affect their preference for a human or AI's input, or their reaction to errors. These findings extend the academic debate about how expertise shapes people's relation to AI. Further, they suggest that new interventions are needed to ensure that people rely appropriately on AI.

Name company or University: Utrecht University

(If applicable) External supervisor: n.a.

UU Supervisor: Talha Özüdoğru

Kontsioti Angeliki - 1659928

Master's Thesis Applied Cognitive Psychology

“The effects of additional use of illustrations on adherence to text-based instructions and rules”

Abstract

The fast pace of modern life requires people to complete tasks efficiently, often by following instructions. This study investigates the cognitive performance of individuals given text-only versus text+illustrations instructions. A sample of 42 participants was divided into two groups: one received text-only instructions, and the other received text+illustrations instructions. The Digit Span Test measured WM capacity, and a Brick Task assessed task performance. Results showed that text-only instructions led to fewer errors compared to text-plus-illustration instructions and completion times were slightly faster for the text-only group. Significant positive correlations were found between task completion time and turnovers, as well as between errors and strict errors. These findings suggest that while illustrations enhance understanding and memory retention, they may also increase cognitive load and error rates during task execution. The study underscores the importance of instructional design and WM in task performance, highlighting potential benefits of WM training for improved adherence to complex instructions. Future research should explore larger samples, varied instructional designs, and the impact of WM training interventions.

Faculty of Social and Behavioural Sciences, University of Utrecht

Dr. Suzanne Hut (Supervisor)

Name: Anna-Maria Paleshnikova, student ID:1670034

Internship:12.5 ECTS

Title: Beyond the Screen: Towards More User-Centric Digital Mental Healthcare

Abstract: In my internship, I looked into a route toward personalizing preventative digital mental healthcare. Ancora Health, the company I worked for, has made significant progress in providing extensive programs for multiple aspects of wellness and lifestyle. What the company is aiming to achieve through scientific research is to provide its users with the most comprehensive care available. However, they face the challenge of delivering personalized care without falling into the 'one-size-fits-all' pitfall of an approach. In an attempt to overcome this, we hypothesized that by clustering the micro-habits in three separate groups ('profiles'), we could then adjust the screening (baseline inventory) to supply us with a better idea of where the user is at the start of their mental health journey and, along with an experienced coach, prioritize correctly the micro-habits so that they have lasting positive results. Despite the time constraints, I believe this could potentially aid the company in laying the groundwork for more user-centered digital mental healthcare.

Company: Ancora Health

UU supervisor: Dr. Anouk Keizer

Company supervisor: Dr. Wanda Wouters, Nurdin Pattipilohy

Name Student: Anna-Maria Paleshnikova (Student ID: 1670034)

Internship or Thesis: Thesis (27.5 ECTs)

Title project: *Beyond the Hype: Does Bionic reading live up to the expectations from the perspective of ADHD individuals?*

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Dr. Suzanne Hut

Abstract

Nowadays, people have nearly unlimited access to a vast amount of written information, and combined with the fast-paced lifestyle led by most, comes the necessity to consume that information as quickly as possible. One of the solutions to the rising necessity to consume information quickly and efficiently is Bionic Reading (BR), which makes quite the claims that it aids in improving speed with minimal trade-off on reading comprehension, especially in individuals with reading difficulties. Bionic Reading is a relatively new method that uses a unique typeface that bolds the initial characters of each word, establishing artificial fixation points that supposedly enhance reading speed while preserving comprehension. The current study aims to evaluate that claim, focusing on individuals with ADHD as they often experience lower processing speed, affecting their reading comprehension. The study will look into whether Bionic Reading can effectively address these challenges. The hypothesis is that Bionic Reading may significantly improve reading speed and comprehension for ADHD individuals, potentially effectively enhancing their understanding of written material.

Name Student: Ioannis Tsoupas

Internship or Thesis: Thesis

Title project: Wayfinding with the use of tactile maps: egocentric and allocentric representations

Name company or University: Utrecht University

UU Supervisor: Dr. Krista E. Overvliet

(Abstract, 100-150 words)

Wayfinding is a complex process interpreted as the ability of an individual to identify their orientation, and to reach any destination quickly and effortlessly. The main processes used during wayfinding are the egocentric and allocentric processes. Despite both strategies being employed during wayfinding, they are not identical. During the egocentric process, a person uses themselves as a point of reference to gain spatial information about an environment. On the other hand, while employing the allocentric strategy, the spatial information is perceived externally; specifically, it focuses on the distance and direction between objects. In this study with the use of tactile maps, I will examine the effects of egocentric and allocentric strategies on route repetition and retracing. Participants will gain information from the tactile maps, and then will have to navigate in a virtual maze. The research question will be examined via employing a 2x2 between subject design.

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Internship or Thesis: Thesis

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Lei Ann Arroccena

Thesis

Mood/Affect and Risky Decision Making

Previous economic crises have shown how emotional responses can exacerbate risk-averse behaviors, leading to suboptimal financial decisions that impact both personal wealth and broader economic stability. By exploring the relationship between affect and risk attitudes, this research aims to provide insights that could enhance financial decision-making processes and inform policies to support individuals in managing their financial risks effectively during periods of economic turbulence. The study involves 150 participants who complete an online questionnaire to assess affective states. In a subsequent lab experiment, participants make a series of decisions involving lotteries with gains and losses. Risk attitudes are measured by plotting utility and probability weighting functions, employing a bisection method to ensure precise elicitation. Key findings indicate that negative affect correlates with increased risk aversion, more pronounced for gains than losses. The results underscore the significant role of affect in financial decision-making, suggesting that mood and anxiety can substantially impact economic behavior.

University of Amsterdam

External Supervisor: Dr. Jan B. Engelmann

UU Supervisor: Dr. David Terburg

Freezing anxiety and defensive mechanisms across threat imminence continuum

This thesis examines defensive behaviors across the threat continuum, focusing on the role of freezing as a common response mechanism among species. The study investigates whether intense freezing behaviors, including startle potentiation and bradycardia, are heightened in individuals reporting higher anxiety characteristics at a personality trait level. The threat continuum concept, encompassing pre-encounter, post-encounter, and circa-strike phases, guides the exploration of defensive responses to varying levels of threat imminence. Drawing on previous research, the thesis delves into cognitive and physiological mechanisms underlying freezing behavior, emphasizing its adaptive significance in navigating potential threats. By examining individual differences in freezing responses and anxiety traits, the study aims to contribute to a deeper understanding of how defensive behaviors manifest across different threat contexts

Name Student: Leonard Schaafsma

Internship or Thesis: Internship

Title project: The effect of oral contraception on fear extinction in an online fear conditioning task

Fear extinction is the process of unlearning a fearful response to a stimulus. As it turns out, the ovarian hormone system in females can be linked to fear related processes, such as fear extinction. And since oral contraceptives, such as the pill, affect female hormones, this could influence an individual's fear extinction. The central research question that will be addressed in this study is the following: "How does oral contraception use affect subjectively measured fear extinction of participants in an online fear conditioning task?". To answer this question, an online fear condition task has been created, which also includes multiple questionnaires.

Name company or University: Utrecht University

(If applicable) External supervisor: n.a.

UU Supervisor: Joke Baas

Name Student: Leonard Schaafsma

Internship or Thesis: Thesis

Title project: How do we perceive touch? An investigation into the nature of the postauricular reflex

The postauricular reflex (PAR) is part of what is known as the startle reflex. Research shows that the PAR is potentiated during appetitive states, such as when viewing pleasant pictures. There hasn't been done much research however into other sensory modalities that could maybe potentiate the PAR. This study will investigate whether receiving a pleasant touch will yield a higher PAR response than viewing pleasant pictures, as well whether a social touch will yield a higher PAR response than a non-social touch. Participants were lightly touched on their arm by one of the researchers, as well as a robot. During the second part of the experiment, participants were exposed to three different categories of pictures (positive, neutral and negative). The results indicate a significant difference in PAR response between viewing pleasant pictures and receiving a pleasant touch. However no significant difference between a social and a non-social touch was found.

Name company or University: Utrecht University

(If applicable) External supervisor: n.a.

UU Supervisor: Anouk Keizer

Lisa Blanckestein

Thesis

Title: **Investigating Tactile Imagery as a Trigger for the Rubber Hand Illusion and exploring correlation.**

Utrecht University

1st supervisor: Dr. Renee Lustenhouwer

2nd supervisor: Dr. Suzanne Hut

Abstract

Our study builds on the rubber hand illusion (RHI) paradigm, investigating the potential induction of the illusion through tactile imagery. Diverging from conventional approaches, we investigate the inducibility of the rubber hand illusion through the mental simulation of experiencing taps on a rubber hand. The project also examines correlations between participants' tactile imagery abilities and the strength of the RHI, offering insights into body representation and multisensory integration. This research holds implications for neurorehabilitation and brain-computer interfaces. In a two-part experiment, participants underwent baseline tactile imagery assessments and engaged in RHI under various conditions. Results suggest a significant positive correlation between tactile imagery abilities and a heightened sense of ownership and proprioceptive drift toward the rubber hand in both classic and tactile imagery-based RHI. Successful induction of RHI through tactile imagery alone emphasizes potential applications in healthcare and technology.

Name Student: Meiqi Lin

Internship

Title project: Rapid Invisible Frequency Tagging in EEG-based Brain Computer Interfaces

Abstract

A Brain-Computer Interface utilizing Steady-State Visual Evoked Potentials (SSVEP) allows people to control external devices, such as computers, by gaining information from human brain activity. This technology enhances the lives of people, particularly those with motor impairments. While most empirical studies in this field focused on the low-frequency tagging technique ($<30\text{Hz}$), recent studies indicate Rapid Invisible Frequency Tagging (RIFT) employing higher frequency ($\geq 60\text{Hz}$) provides more opportunities for cognitive research and offers potential improvements in user experience and performance. The present study aimed to test the feasibility of tagging stimuli with multiple invisible frequencies to implement Brain-Computer Interfaces with electroencephalography (EEG) data. During the experiment, a grid of nine numbers (1-9) was displayed on the screen, with each number flickering at a distinct frequency ranging from 60-68 Hz, with an interval of 1 Hz. Participants were instructed to attend to the target number on each trial, allowing the classifier to discern their intentions by EEG data. The results revealed a notably low classification accuracy, indicating the classifier was unable to decode specific flickering frequencies. Consequently, the application of RIFT to EEG-based BCI using multiple invisible frequencies simultaneously is unfeasible.

Keywords: brain-computer interface (BCI); electroencephalography (EEG); rapid invisible frequency tagging (RIFT); Steady-State Visual Evoked Potentials (SSVEP)

Name University: Utrecht University

UU Supervisor: Samson Chota

Name Student: Meiqi Lin

Thesis

Title project: The relationship between spatial imagery ability and response inhibition

Abstract

This study is to investigate the relationship between spatial imagery ability and response inhibition. Response inhibition, as the center of executive control, refers to suppressing the irrelevant or inappropriate actions. It is often investigated by the stop-signal paradigm in the laboratory. While extensive studies have indicated the association between motor imagery and response inhibition, there is a notable gap when it comes to the relationship between spatial imagery and response inhibition. By correlation and regression analysis, this project is aimed to test whether spatial imagery ability related to stop-signal task performance and whether spatial imagery can be a predictor of response inhibition. Our results failed to demonstrate the association between spatial imagery and Stop-Signal Reaction Time (SSRT), but we proved that spatial imagery ability is a significant predictor of omissions in Stop-Signal Task.

Keywords: cognitive control; response inhibition; stop-signal task; spatial imagery ability; mental imagery

Name University: Utrecht University

UU Supervisor: Daan van Rooij

Oral Perception: Individual differences in tactile and proprioceptive acuity of the mouth

This research addresses the limited exploration of spatial acuity in the mouth despite its multifunctional use. We set out to explore more about the field of oral perception in terms of tactile and proprioceptive acuity, using oral numeration and shape recognition as the basis. Additionally, as fungiform papillae play an important part in increasing sensitivity of tasting ability, we hypothesized that individuals with heightened taste acuity would perform better in oral spatial acuity tasks and henceforth individuals who perform better in oral numeration would also perform better in shape recognition. Additionally, the study delves the differences between the counting of smaller and larger elements, along with subitizing. We aimed to understand the correlations among oral numeration, oral shape recognition, and the ability to taste PROP substance. To do this, we employed a quantitative within-participants experimental design with multiple tasks. Surprisingly, the results reveal a statistically significant correlation between non-tasters of the PROP substance and superior performance in oral numeration and shape recognition tasks, contradicting our initial hypothesis. Furthermore, a close positive relationship is observed between performance in individuals' numeration and shape recognition tasks, as expected. Lastly, following principles of visual numeration, participants also consistently exhibit a tendency to underestimate smaller elements and overestimate larger ones during oral numeration.

Name Student: Erin Mino

Internship

Title project: Perception of emotion from facial expressions in paintings from 13th to 19th century

Abstract:

The perception of emotion in paintings is a key aspect of understanding how art communicates human experience. This study examines the perception of emotion from facial expressions in paintings from the 13th to the 19th century. We aim to explore changes based on art style and historical period, as well as investigate the relationship between painting intensity and facial expression intensity. In a pilot study with 15 participants, we analyzed 20 paintings focused on "Nativity" (Birth of Christ) and "Pieta" (Death of Christ) themes. Participants matched the faces in the paintings to 8 basic emotions. Results revealed low overall agreeability but higher consensus for "Pieta" paintings. Based on these findings, the main study expands to 56 paintings and 150 participants, with refined instructions. This larger study aims to assess how well people perceive emotions across different periods and styles, and whether the expressiveness of a painting correlates with the intensity of facial expressions.

University:

TU DELFT

External supervisor:

Maarten Wijntjes

UU Supervisor:

Chris Janssen

Name Student: Marta del Rosario Perez Barrios

Internship or Thesis: Internship

Title project: Creating an Experiment to Overcome Limitations in Computational Aesthetics

From an objectivist point of view, it has been found that certain visual characteristics of an image have the potential to influence aesthetic preferences, with previous research suggesting that these key visual characteristics, also called image statistics, vary depending on rating scale and artistic genre. However, most of these studies have overlooked individual differences by employing mean ratings averaged from different groups of participants. To address this limitation, I designed and ran an experiment ($n=10$) where a consistent rater evaluated their experience of a dataset of painted art images in the scales of 'beauty', 'liking', 'emotionality' and 'appeal'. This dataset was created by retrieving 960 landscape, portrait, still-life and abstract artworks from WikiArt.org and the DELAUNAY dataset. Expected results would highlight the importance of analyzing results on an individual level, and would support an interactionist view on computational aesthetics.

Name company or University: Utrecht University

UU Supervisor: Sjoerd Stuit

Name Student: Marta del Rosario Perez Barrios

Thesis

Title project: How do adults with a history of childhood trauma respond to threat on different levels of imminence?

This thesis examines how adults respond to threats at varying levels of imminence and the influence of childhood trauma on these responses. Using the threat imminence continuum model, it investigates defensive behaviors such as freezing, startle reflex, heart rate, and skin conductance. Freezing is expected to be highest during distal threats and decrease as the threat becomes imminent, transitioning to active fight or flight responses. Participants underwent a threat of shock task while their physiological responses were recorded using a BioSemi system and a custom balance board. The study hypothesizes that individuals with childhood trauma will exhibit stronger physiological responses across all threat levels, with heightened freezing during distal threats and increased heart rate and skin conductance during imminent threats. This research aims to gain a deeper understanding of how early trauma influences defensive behaviors and physiological threat responses.

Utrecht University

UU Supervisor: Joke Baas

Name Student: Naomi ter Haar

Internship or Thesis: Internship (Combined)

Title project: UX in a Work Setting: “Creating a User-Experience Measurement Tool for Nedap’s Ons Suite”

The aim of this study is to define how to quantitatively assess the user experience of Nedap’s Ons Suite, a work-tool designed to simplify the administrative tasks of healthcare workers. Previous work on User-Experience has resulted in an extensive number of measurement tools, each focusing on different factors of the experience but all in a leisure context. Using the Critical Experience Method, this study focused on what the end-users of Ons Suite deemed the most important factors in determining their User-Experience. Results showed similar experiences and preferences for different types of end-users, with themes like a big sense of responsibility and a need for a reliable, efficient and trustworthy work-tool. This implies one measurement tool would be suitable to measure the experience of all end-users. Interpretation of the results was translated to a measurement tool consisting of a semantic differential questionnaire that measured nine UX factors centred around the themes Usability and Reliability.

Name company or university: Nedap

External supervisor: Lex van Velsen

UU supervisor: Martijn Mulder

Name Student: Naomi ter Haar

Internship or Thesis: Internship (Combined)

Title project: UX in a Work Setting: “Creating a User-Experience Measurement Tool for Nedap’s Ons Suite”

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Name company or university: Nedap

External supervisor: Lex van Velsen

UU supervisor: Martijn Mulder

Name student: Naomi ter Haar

Internship or Thesis: Thesis (combined)

Title Project: Validating the ONStool: “a User-Experience Measurement Tool for Electronic Client Dossier Systems”

ABSTRACT

The aim of this study is to verify the reliability and validity of the ONStool: a measurement tool designed to quantitatively assess the user experience of ECD systems. To validate the tool a mixed methods approach was applied. We conducted a threefold experiment that included the ONStool, the SUS questionnaire and a word association exercise. With the results of the experiment the Construct Validity, Internal Validity and reliability of the tool were determined. Despite the factor analysis revealing two distinct factors, we argue the tool is unidimensional based on the holistic nature of UX. For the internal validity, a strong correlation was found with the SUS questionnaire, which supports the assumption that part of the ONStool measures the theme usability. As for the reliability, a value of .917 was found for Cronbach’s alpha, indicating a very high internal consistency.

Name company: Nedap Technology

External supervisor: Lex van Velsen

UU supervisor: Martijn Mulder

Name Student: Mina Wali

Internship or Thesis: Thesis

Title project: How do people use their senses when navigating?

Spatial navigation, a crucial cognitive function, involves wayfinding, the task of reaching a target location using environmental cues.

This study investigates wayfinding strategies, such as, survey based, egocentric and procedural wayfinding strategies and their efficacy within a virtual maze setting. Participants were asked to play a virtual maze game with 4 levels (conditions) and were then asked to complete a questionnaire on their self-reported preferred way finding strategies.

The conditions consisted of, auditory landmarks, haptic landmarks, audio haptic landmarks and a control condition with no landmarks. The completion times were recorded. Results demonstrate that multimodal (audio-haptic) landmarks facilitated the shortest completion times, suggesting their superiority over unimodal landmarks. This study underscores the nuanced nature of navigation and advocates for further research examining multimodal landmarks across diverse environments.

Name company or University: Utrecht University

UU Supervisor: Dominique Blokland

Name Student: Mina Wali

Thesis

Title project: Does theta power, prior to a go-signal, predict successful inhibition in the SST?

Abstract:

Cognitive control, essential for functions like attention and response inhibition, is known to be impaired by frontal lobe damage, leading to distractibility, perseveration, and impulsiveness. Theta oscillations in the frontal cortex are implicated in cognitive control, but their role in response inhibition remains unclear. This study explores the relationship between theta power and response inhibition during a stop-signal task (SST). Prior research suggests increased theta power may indicate proactive cognitive control. We hypothesized that higher theta power would be observed before successful inhibition trials compared to failed ones. We replicated van Rooij et al (2024), which suggested a link between lower frontal theta power and difficulty in response inhibition. Using EEG, we measured theta power in 59 participants (aged 18-25) while they performed the SST. Our findings aim to provide further evidence for theta power's role in proactive response inhibition, contributing to the understanding of neural mechanisms underlying cognitive control.

Location : Utrecht University

UU Supervisor: Sam Van Bijnen

Name Student: Leann Leung

Internship or Thesis: Internship

Title project: Behavioural-Metrics for Authentication For Navy Combat Information Center – A Feasibility Study

The objective of this study is to explore the feasibility of applying validated continuous, unobtrusive, and multi-modal behavioural metrics for authenticating users in the naval control room setting. The research question is: *To what extent is the integration of behavioural metrics a viable approach for authenticating users of the Multi-Operator Console in the Naval Combat Information Center?* Individual behavioural metrics such as keystroke dynamics, mouse dynamics, speaker detection and eye-tracking, as well as multi-modal behavioural metrics such as combining keystroke and mouse dynamics, and combining mouse dynamics and eye-tracking are explored. Feasibility criteria such as metric performance, practicality (cost, data required), and privacy concerns are considered. A preliminary review shows that keystroke dynamics is feasible for the target setting due to its accuracy and low implementation cost, with the combination of keystroke dynamics and mouse dynamics area especially feasible for uncontrolled tasks in dynamic operational settings.

Name company or University: Thales Nederland B.V.

External supervisor: Dr. Iris Cohen

UU Supervisor: Dr. Sam van Bijnen

Thesis Abstract for ACP January Update Day

Name Student: Leann Leung (2669757)

Internship or Thesis: Thesis

Title project: Virtuality Check: A comparative analysis of virtual reality and mock-ups to enhance comprehension of novel maritime operational concepts

Name company or University: Maritime Research Institute Netherlands (MARIN)

External supervisor: Heike Diepeveen (H.Diepeveen@marin.nl)

UU Supervisor: Chris Klink (p.c.klink@uu.nl)

Abstract:

The maritime industry faces the challenge of limited end-user involvement in ship designs. This study explores various method arrangements to aid maritime workers to visualize and understand a novel operational concept, and determines the most optimal method arrangement for enhancing novel concept comprehension. Methods to be tested include high-fidelity virtual reality, low-fidelity mock-up exercise, and a combination of both methods. 24 participants from the Dutch maritime sector first complete a pre-test task of constructing a concept map with minimum written information about a novel remote navigation scenario. Then they experience the full scenario via their assigned method(s). Finally, they undergo post-test questionnaires to measure immersion and cognitive engagement, and construct concept maps again to evaluate their understanding of the scenario. The results of this study will illuminate the most promising arrangement of methods that balances realism and practicality to support novel concept imagination from the ideation to the prototyping stage in the concept development process.

Name Student: Alfred Norén

Internship or Thesis: Internship

Title project: Perception Study on the Noise Impact of New Generation Aircraft Types

Aircraft noise significantly impacts residents near major airports, affecting their quality of life and health. The aim of this internship is to design and conduct a perception study to evaluate the noise annoyance caused by two new-generation aircraft, the Airbus A320-NEO and Boeing 737-Max, compared to older models. Noise annoyance is a complex response that depends on acoustic factors, such as loudness and specific frequencies, and non-acoustic factors like noise sensitivity and coping capacity. Using the Multiple-Item Annoyance Scale (MIAS), the research assesses participants' annoyance responses to standardized recordings of various aircraft types, as well as train noise for additional comparison. Participants listen to the sound recordings through headphones while performing a reading task, incorporating task interference into the annoyance assessment. The outcomes aim to improve noise mitigation strategies and inform communities about the noise characteristics of newer aircraft models.

Name company or University: NLR

(If applicable) External supervisor: Naomi Sieben

UU Supervisor: Chris Klink

Name Student: Ece Çavuşoğlu

Internship or Thesis: Thesis

Evaluation of an App-Based Healthcare Intervention for the Prevention of Sarcopenic Obesity in Older Adults

Abstract

The primary goal of this thesis project is to assess the user experience of a mobile application, SO-NUTS. SO-NUTS is a mobile health and fitness application targeted towards retirement age individuals to encourage them to adopt a healthy diet and engage in physical activity regularly for the prevention of sarcopenic obesity. Given that it is a recently developed, novel mobile app, it is crucial to evaluate the user experience in order to predict user satisfaction and to alter and enhance the design features. The participants were asked to download, use the app according to the instructions and then complete an online questionnaire to assess the usability, their technology acceptance and satisfaction regarding the app. The analysis of quantitative data from the self-report questionnaire indicated how usable the app is and measured user satisfaction and acceptance. The qualitative data from open-ended questions were analyzed through thematic coding to pinpoint the specific topics that needs to be addressed. The findings can inform the designers of the app about how users experience the app and offer ways to improve the app according to the user feedbacks.

Keywords: health, fitness, nutrition, mobile app, user experience, usability, technology acceptance

Name company or University: University of Amsterdam and Amsterdam University of Applied Sciences

External supervisor: Prof. Somaya Ben Allouch

UU Supervisor: Prof. Joana Barbosa Vieira

Abstract

Background

Human interaction affects emotional health, cognitive development, and social learning. Physiological synchrony, where individuals' physiological processes mirror each other during interactions, is especially vital in parent-child relationships for fostering emotional bonding and empathy. However, analyzing heart rate variability (HRV) to understand interaction states presents challenges, particularly with neurodivergent children who show atypical responses. This study utilizes one-dimensional convolutional neural networks (1D CNNs) to enhance the analysis of these complex signals.

Methods

This study employs 1D CNNs to classify interaction states using HRV data from neurodivergent child-caregiver dyads. The approach combines hand-crafted feature extraction with supervised learning, optimizing neural network architectures to process time-series data efficiently. The model's performance is evaluated using accuracy and F1 score metrics. The neural network architecture includes layers for convolution, ReLU activation, and dense layers, concluding with a softmax output layer for classification.

Results

The neural network achieved 97% for accuracy and F1 score in ternary classification of interaction states. The results highlight the effectiveness of using deep learning for HRV analysis in this context.

Conclusions

The study demonstrated the potential of deep neural networks to develop robust, continuous, and noninvasive methods for synchrony detection. The ultimate goal is to improve the quality of life for neurodivergent children and their caregivers through enhanced understanding.

Name Student: Thelma Solleveld

Internship or Thesis: Internship

Title project: Points of attention in Human Factors analyses of the pedestrian task by Rijkswaterstaat

Rijkswaterstaat performs trip analyses using five human factors principles —expectations, perception, comprehensibility, capability, and willingness—to analyze the interaction of the characteristics of the user and the environment and the influence of this on their behavior. This is focused on highways and waterways and a specific method for pedestrian paths lacks. This study identifies points of attention for human factors analyses of the pedestrian task. A database of prior cases was filtered to select six cases with expert observations. These observations were coded to identify common elements and checked through an expert feedback session. The study found that pedestrians frequently use paths in the opposite direction; it is important that there is a (clear) separation from the cycling path; it is important that the impression a location gives matches the wanted behavior; it is important that people can recognize what the moving part of the bridge is; and that it is important to consider different kinds of conditions and different kinds of pedestrians. Problems and risks concerning the pedestrian task can be discovered by taking these points into account during the analyses. This way, better and more fitting advices can be given concerning pedestrians.

Name company or University: Rijkswaterstaat

(If applicable) External supervisor: Anke Mulder

UU Supervisor: Chris Janssen

Name Student: Thelma Solleveld

Internship or Thesis: Thesis

Title project: The rubber hand illusion through tactile imagery

Abstract:

The study investigates whether tactile imagery can induce the rubber hand illusion, a phenomenon where people perceive a rubber hand as their own through synchronized tactile and visual stimuli. Participants engaged in synchronous and asynchronous tactile imagery while observing a rubber hand being touched. Results showed that synchronous tactile imagery led to a significant shift in perceived hand location towards the rubber hand. However, feelings of ownership did not significantly differ between synchronous and asynchronous imagery conditions. This dissociation between proprioceptive drift and ownership suggests that different mechanisms could be underlying them. This study underscores the complexity of multisensory integration and highlights potential applications in neurorehabilitation and artificial limbs. Future research should investigate individual differences in tactile imagery ability and refine experimental procedures to further explore the possibilities of tactile imagery in the rubber hand illusion.

Name company or University: Universiteit Utrecht

(If applicable) External supervisor: /

UU Supervisor: Renee Lustenhouwer

Descriptive Abstract

Name: Hyunseo Cho

Internship or Thesis: Thesis (27,5 ECTS)

Title: The Association between Cognitive Complaints in Daily Life and Passive Coping Style of Healthy Individuals in South Korea

Cognitive complaints are self-reported issues reflecting individuals' cognitive impairment in daily life, affecting activities such as work and social interactions. These complaints often go undetected by neuropsychological tests, which may not capture the subjective cognitive deterioration experienced in day-to-day life. Recognizing the need for more sensitive assessments, standardized tools (e.g., CoCo-P) have been developed to measure cognitive complaints across domains like executive function, memory, and attention. Research highlights an association between cognitive complaints and passive coping, particularly among patients with acquired brain injuries. However, this relationship is underexplored in healthy individuals. In South Korea's competitive society, cognitive burdens are prevalent, necessitating an understanding of how passive coping impacts cognitive complaints. Thus, this thesis aims to explore the relationship between daily cognitive complaints and passive coping styles among healthy South Koreans using the Cognitive Complaint – Participation (CoCo-P) assessment, focusing on examining cognitive complaint scores and their association with passive coping styles.

Name company or University: Utrecht University

UU Supervisor: Eileen Bousché

Name Student: Nurfitriani Marsuki

Internship or Thesis :Thesis (27.5 ECTS)

Title project : Effects of Verbal and Visual Working Memory Capacity on Reinforcement Learning Performance and Strategy Use

This study investigates the influence of verbal and visual working memory capacity (WMC) on reinforcement learning (RL) performance and strategy use in tasks involving concrete and abstract stimuli. We assessed general, verbal, and visual WMC and found significant correlations among these capacities. However, only visual WMC showed a significant correlation with RL performance. The type of stimuli presented had a notable impact on learning accuracy, with concrete stimuli resulting in superior performance compared to abstract stimuli. Participants predominantly employed chunking strategies to manage cognitive load during tasks. The findings highlight the specific contribution of visual WMC to RL and emphasize the importance of stimulus type in learning processes. Future research should aim to refine the assessment of WMC and explore further the domain-specific influences on RL.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Dr. Irene van de Vijver

Name Student: Nurfitriani Marsuki

Internship or Thesis: Internship (7.5 ECTS)

Title project: Haptic and Auditory Enumeration of the Contents of a Container

(Abstract, 100-150 words)

Gaining information from surroundings may involve multiple senses. The more sources people have, the more robust their decisions will be. In this study, we examine how participants estimate the number of items in the containers ($n = 1$ to 8) under three different conditions: haptic only, auditory only, and haptic-auditory, and we recorded the reaction time. A total of 18 participants were included, each had all three conditions. Participants were instructed to guess the number of items inside the containers by shaking them. Results showed that sensory modalities significantly influenced both estimation accuracy and the number of items. The multisensory modality (combination haptic and auditory) performed better than auditory-only condition because people have more information in the haptic-auditory condition. Furthermore, sensory modalities also had a significant effect on the reaction time which varied according to the number of items, it means the fewer items in the container, the shorter the reaction time.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Dr. Krista Overvliet

Name Student: Yuye Yang

Internship or Thesis: Thesis

Title project: Exploring HRV synchrony Parents and Children with ASD during Stressful Interactions

Abstract

This study explores the correlation of heart rate variability (HRV) between parents and children with Autism Spectrum Disorder (ASD) under various stress conditions. We hypothesized that there is a synchrony of HRV between parents and their ASD children under various stress conditions. To investigate this, we are trying to design a stressful interaction experiment with varying stress conditions and quantitatively measure both the physical responses of ASD children and their parents by employing wearable HRV monitors. The purpose of this study is to gain an understanding of emotion regulation when they face different stressful interaction activities. These outcomes have profound implications for the emotional dynamics of families affected by ASD. Also, the results of this research will provide potential insights for developing some treatments for ASD children.

Name company or University: TU Eindhoven

(If applicable) External supervisor: Jing Li and Emilia Barakova

UU Supervisor: Suzanne Hut

Name student: Yuye Yang

Internship or Thesis: Thesis

Title project: Exploring the Correlation between Stress-related Heart Rate Variability (HRV) in both Parents and Children with ASD and their subjective stress levels

Abstract:

This study aimed to examine the association between subjective psychological stress and objective stress-related HRV in both parents and children with Autism Spectrum Disorder(ASD) under different interaction contexts. We hypothesized that there is a correlation between subjective psychological stress and objective stress-related HRV in both parents and children with ASD, regardless of the stress situations. To investigate this, we designed an about one-hour behavior experiment that include stressor for children, stressor for parents, and stressor for both parents and children. We collected parents and children with ASD's HRV during the experiment, and analysis their heart rate data thorough Kubios, which is an accurate sofeware to preprocess ECG signal. To get more details, we chose 30s as our window width.

University: TU Eindhoven

External Supervisor: Jing Li and Barakova

Internal Supervisor: Christoph Strauch

Name student: Tanja van der Marel

Internship or Thesis: Thesis

Title project: *Impact Corona Pandemic on Youth Mental Health*

The corona crisis seems to have reinforced the existing downward trend in mental health among young people. At the same time, much remains unclear about the effects of the corona crisis on youth mental health; it is not known how the observed deterioration fits into a broader trend picture including pre- and post-pandemic period, and what factors play a role in this deterioration to a greater or lesser extent. Anxiety symptoms and depressive symptoms play an important role in predicting future mental health problems. For this reason, the thesis report will focus on these symptoms and will try to answer the following research question: Did the corona pandemic disrupt or accelerate existing trends in the prevalence of anxiety and depression symptoms among youth worldwide? And if so, what could be advised to policymakers to mitigate this impact and prevent such impact for following pandemics?

Name company or University: Rijksinstituut voor Volksgezondheid en Milieu (RIVM) & Nivel

(If applicable) External supervisor: Marjonneke de Vetten-Mc Mahon & Mark Bosmans

UU Supervisor: Suzanne Hut

Name Student: Tanja van der Marel

Internship or Thesis: Internship

Title project: Ontwikkeling Vragenbibliotheek Belevingsonderzoek

The aim of this internship is to construct a library filled with experience questions, that will facilitate setting up a questionnaire at Goudappel. Constructing a questionnaire at the mobility company is often time consuming, since many theories and guidelines have to be taken into account. For this reason, the library was developed by the use of PowerBI. The data of the library originated from a collection in an Excel file, consisting of all questions used in experience research at Goudappel and partner companies from the years 2016 up until 2023. A strict selection was made from the collection, which led to a list of high-quality questions. The interface consisted of a filter panel and a table; the list of questions in the table alters when filters are applied in the filter panel. Employees at Goudappel can now use the library to find relevant questions that one needs for their experience research. For future recommendations, an automated interface can be developed, which produces an instant ready-to-use questionnaire, instead of a list of possible questions that can be used for one's experience research.

Name company or University: Goudappel

(If applicable) External supervisor: Daphne Rutgers & Rick Schotman

UU Supervisor: Gabriël Beckers

Name: Julia Ganama

Project Title: Influence of Covert Attention on Pupil Size Change with Complex Stimuli

Course: Thesis

Abstract:

Covert attention, the ability to focus on peripheral areas without making eye movements, is categorized into endogenous (top-down) and exogenous (bottom-up) types. This study examines how pupil size changes can reveal the deployment of covert attention using the Open-DPSM toolkit, which models pupil responses to dynamic stimuli. Thirty-eight participants viewed 32 movie clips under varied attentional conditions and tasks. The results showed a significant horizontal pupil bias when comparing left versus right attentional conditions, indicating that pupil size changes can reflect endogenous covert attention shifts. Additionally, participants watched either a mirrored or original set of movies, revealing the impact of low-level features on pupil size changes. A significant correlation between pupil bias and contrast event bias was found, highlighting the role of exogenous covert attention and low-level properties.

University: Utrecht University

Supervisor: Christoph Strauch

Name Student: Beinu (Vivian) Xu

Internship or Thesis: Thesis

Title project: The influence of subtitle design of short videos on people's perception

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Chris Klink

Abstract

My thesis is on how people's perceptions on short videos can be influenced by the setup of subtitles. As short videos are being viral, and the users of short-video platform are expanding dramatically. The way people perceive videos has changed. Besides the things that the videos turn from horizontal to vertical to fit the average use habit of smartphones among young people, people are exposed to more irrelevant stimuli for a longer period. Meanwhile, the usage of subtitles is also very different from the traditional ones. Therefore, I designed an experiment with videos with different subtitles set up and asking participants questions to see their perceptions on the video. The fixations in certain ROI are also investigated to study their visual attention distribution.

*The link of the form is not available so I stay with this one.

Name Student: Beinuo(Vivian) Xu

Internship or Thesis: Internship

Title project: My internship aimed to develop a viable experimental setup for studying visual attention in dyadic interactions using eye-tracking devices and vision-based fiducial markers, such as ArUco markers.

The primary learning objects addressed in my internship were: (1) the characteristics of eye-tracking devices, (2) methods for studying dyadic visual attention, (3) accurately placing fixations from two participants on the same reference frame, and (4) interpreting gaze data.

During the internship, various approaches were explored to align the gaze of two participants on a reference frame using ArUco markers and Apriltags, both being vision-based fiducial marker pose estimation tools. As a result, the experimental setup was finalized, enabling the study of joint visual attention in scenarios where both participants wear Pupil Labs Neon and view teaching materials surrounded by ArUco markers on a screen.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Margot van Wermeskerken & Roy Hessels

Name of Student: Marina Misioni

Thesis Project

Title project: Threat Imminence & Prosociality

Abstract

This thesis investigates how the immediacy of threats influences prosocial behavior. The study examines whether individuals are more likely to help others when faced with immediate (imminent) versus anticipated (distal) threats. Using a within-subjects experimental design, participants will be exposed to varying threat levels while their physiological responses (heart rate, skin conductance, and balance) and decisions to either escape or help others are measured. The main hypotheses are that imminent threats will increase prosocial behavior due to heightened physiological arousal. This research uses neuroimaging and behavioral tests to uncover the mechanisms driving these behaviors. The findings aim to clarify how threat perception affects prosocial actions, providing insights into human responses to danger. This study's outcomes could have significant implications for understanding and improving prosocial behavior in high-stress environments, with potential applications in therapy and crisis management.

Utrecht University

UU Supervisor: Joke Baas & Joana Barbosa Vieira

Name Student: Julianna Smolenski (6098290)

Thesis

Title project: The Relationship Between Trauma and Stress as Influences of Cognitive Control

Abstract: The purpose of this study is to investigate the relationship between stress as a factor of trauma and cognitive control. Using EEG spectral analysis and cortical asymmetry calculation, I aim to identify the neural correlates of childhood trauma and stress, particularly as they influence cognitive control in a non-clinical population. The study included healthy participants with varying stress levels and exposure to childhood trauma, as measured by the Perceived Stress Scale (PSS) and the Childhood Trauma Questionnaire (CTQ), and consisted of a stop task with resting state measures taken both before and after task trials. In the analysis, I am comparing CTQ and PSS Scores to alpha measures during resting state, using frontal asymmetry to represent cognitive control. The findings of this study may contribute to existing and future research on the effects of trauma and stress on cognitive functioning.

Utrecht University

UU Supervisor: Leon Kenemans

Name Student: Zehra Öztürk

Internship or Thesis: Thesis

Title project: An exploratory study about human motion perception: The ocular–eccentricity induced motion illusion and its estimated velocity.

This study discusses the presence and perceived velocity of a visual illusion introduced as the ‘ocular eccentricity induced motion illusion’ (MI), in which a stationary target is falsely perceived to move eccentrically into a horizontally fixated eye-direction while the head, eyes and environment are kept stationary. A relevant theoretical background in relation to human motion perception is described. Results indicated that MI was present ($p = 0.002$, $\alpha = 0.05$) and had a proportionally larger effect when the target was positioned further from the middle of the visual frame. Furthermore, results indicated that participants saw MI at a velocity of $0.1^\circ/\text{s}$ when the visual target was positioned at 9.9° to the left or right of the straight-ahead orientation of the head. These findings may strengthen the theoretical background for implications in treatment for vestibular disorders and aviation safety. Overall, this study develops our understanding of basic human motion perception.

Name company or University: Utrecht University and Vrije Universiteit

External supervisors: Jelte Bos and Jeroen Smeets

UU Supervisor: Maarten van der Smagt

Boris Ahsmann

Internship

An evaluation of Spoofy's educational capabilities

Abstract

For this internship, I investigated the educational capabilities of the serious game 'Spoofy', a game about cybersecurity intended for children from approximately 7 to 12 years old. Testing with the target audience was only done during development and not after, only with the Finnish version. I studied the Dutch version of Spoofy. This research is aimed at testing how effective Spoofy is as an educational tool. I conducted experimental research to gather quantitative data from students from five classes from three primary schools with children ranging from 6 to 13 years old. They were asked to play a level from Spoofy and a control game with varied order. Before and after playing each game, the participants were presented with 3 scenarios and a Likert scale with emojis to assign a sentiment. Data is still being gathered, but if Spoofy is effective, sentiments will change in a manner that aligns with Spoofy's lessons after playing Spoofy and not after playing the control game.

Company name: CGI

External supervisor: Chantal Achterberg

UU Supervisor: Surya Gayet

Boris Ahsmann

Thesis

The acceptance of POS systems in the retail sector

Abstract

For this project, I investigated the acceptance of point-of-sales (POS) systems in retail. POS systems are all-in-one sales systems for retail stores. Employees sometimes have difficulty adapting to new systems like these. So, the aim of this research is to find out whether Technology Acceptance models can be used to predict how well these POS systems are accepted by employees in the retail sector. Using the Unified Theory of Acceptance and Use of Technology (UTAUT) and the Task-Technology Fit (TTF) models, I am conducting qualitative research through semi-structured interviews with retail employees from stores in the Randstad area. This data is currently being gathered. When completed, this research could give insight into the use of POS in the retail sector and what psychological factors influence the acceptance of these systems. The findings aim to provide actionable recommendations for improving future POS system implementations.

Company name: CGI

External supervisor: Tom Kleiberg

UU Supervisor: Leendert van Maanen

Abstract poster presentation

Name Student: Julie Heijnen (6421245)

Internship or Thesis: Internship

Title project: Customer experience research for a.s.r. Income

Abstract:

a.s.r., a leading Dutch insurance company, offers a comprehensive range of financial products. During my UX/CX internship in the Delivery Team within the Income department of a.s.r., I conducted interviews with employees to investigate their experiences and organized the findings. I validated the data in follow-up meetings. Besides that, I identified key objectives, formulated recommendations, and collaborated with colleagues to deliver a proposal to the Management Team. The objective of this study was to determine whether Microsoft Dynamics 365 could enhance the efficiency, effectiveness, and enjoyment of work processes for a.s.r. Income employees, thereby improving a.s.r.'s customer experience. Currently, employees face challenges integrating various tools, which negatively impacts customer experience. The recommendation is to implement Microsoft Dynamics 365 to streamline work processes, reducing time consumption, errors, and frustration. This implementation is expected to enhance efficiency, effectiveness, and job satisfaction, thereby improving the customer experience for a.s.r. Income.

Name company or University: a.s.r. verzekeringen

(If applicable) External supervisor: Kiki Maes (kiki.maes@asr.nl)

UU Supervisor: Irene van de Vijver

Name Student: Koert Stribos
Internship or Thesis: Internship
Title project: Snip-and-Stitch:
A Simple, Universal Correction for the Pupil Foreshortening Error

The Pupil Foreshortening Error (PFE) causes an effect of gaze position on pupil size readings, which requires researchers to heavily limit their design or implement complex corrections. We here introduce a novel, simple correction method which ‘snips’ the pupil size before a saccade, and ‘stitches’ the pupil size back after the saccade, which disregards the change in pupil size caused by the saccade. A simple task was used for the implementation of this method. By applying the snip-and-stitch to the resulting dataset, error in reported pupil size was reduced by 65%. Overall, the snip-and-stitch is not as effective in reducing error in reported pupil size as existing methods, but the snip-and-stitch does offer an advantage in that it can be applied post-hoc, and that it theoretically works for all types of eye-trackers.

Name company or University: Utrecht University
UU Supervisor: Christoph Strauch

Name Student: Macy Bouwhuizen

Internship or Thesis: Internship

Title project: Order of Elements on Complex Websites for Screen Reader Users

Navigating complex websites presents significant challenges for screen reader users due to non-linear content presentation. This study investigates the optimal order of webpage elements to enhance accessibility and usability for visually impaired (VI) individuals using screen readers. Conducted with seven blind participants, the research involved three phases: exploring simulated product pages, ranking element combinations, and evaluating an optimized product page. Results indicate that price information, product title, and product images with ALT-text are the most critical elements. These findings suggest prioritizing these elements in web design to improve the online experience for screen reader users, addressing common frustrations such as inaccessible content and confusing layouts. This research contributes to better design practices, fostering greater inclusivity for the VI population online.

Name company or University: Accessibility Foundation

(If applicable) External supervisor: Veron Wormeester

UU Supervisor: Krista Overvliet

Name Student: Macy Bouwhuizen

Internship or Thesis: Thesis

Title project: The role of strategy selection in economic decision making focused on reaction time.

When individuals are presented with varying amounts of money, each associated with different probabilities of winning, some choose intuitively, while others choose rationally. Within the framework of dual-process theory, we examined how participants develop strategies and adjust them based on feedback. Overall, the research aimed to bridge gaps in understanding the determinants of economic decision-making. We employed an online experimental setup for two experiments. In the first experiment, we demonstrated that participants develop a certain strategy, which we modeled based on reaction time and self-reports into a Hidden Markov Model. In the second experiment, we manipulated the order in which participants received feedback and showed that while some individuals could be nudged toward certain strategies (e.g., safe or risky decisions), others remained consistent with their original strategy. In conclusion, we found significant individual variability within this framework, indicating that economic decision-making strategies and their transitions differ among people.

Name company or University: Utrecht University

UU Supervisor: Leendert van Maanen

Abstract Updated day 3

Name: Ruben van der Linden

Internship: Reviewing the research methods used within the human factors department at Marin.

Abstract:

MARIN is currently working with Noldus to develop an integrated data acquisition platform capable of reading, processing and analyzing data from multiple sensors. As part of this, MARIN requested that I investigate what is currently possible with the sensors that they already have available as well as some sensors that they might want to use in the future. To do this I conducted a non-systematic literature study, where I compiled the current state of human factors research in the fields of eyetracking, heart rate variability, and electrodermal activity. The final result is a report including this information as well as recommendations on the use of these measures and other measures that might be interesting to investigate in the future. Finally I identified three main trends that describe the direction that human factors research is taking and how they relate to each other.

Company: MARIN

External supervisor: Niklas van Duinen (K.W.Duinen@Marin.nl)

UU Supervisor: Chris Klink (p.c.klink@uu.nl)

Ruben van der Linden

Thesis

Modelling the cognitive effects of military operations

Abstract:

The process by which we form decisions is highly complex. During high stress situations these decision can be compromised due to lack of information, stress or emotion. Human factors is showing an ever growing interest in the early evaluation of operator state and the way that changes their decision-making. In my thesis I will be focusing on the measure of pupil size to identify uncertainty during decision formation and the effect decision-making. By this we hope to identify early signals of uncertainty of operators. Literature research so far shows promising signals for being able to do this and I am currently working on formulating the methods for this research.

Auditor: NLR, Carmen van Klaren

Assessor: Samson Chota

Name Student: Ilona van Gestel

Internship or Thesis: Thesis

Title project: The effect of Bionic Reading on the reading comprehension of Dyslexics

Dyslexics often show difficulties in the reading process. The designer of the Bionic Reading (BR) font claims that it should support the reading flow, improving reading speed and comprehension for especially individuals with dyslexia. This study therefore investigated the reading comprehension, focussing on the N400 component, in dyslexic readers compared to typical readers regarding the different fonts. EEG of both groups was recorded while reading Dutch congruent and incongruent sentences displayed in BR or standard font (SF), utilising rapid serial visual presentation (RSVP). As an attentional check, questions were asked in approximately a quarter of the sentences. A 6-way repeated measures ANOVA revealed that incongruent stimuli elicited a stronger N400 response compared to congruent stimuli, where typical readers exhibited a significant N400 effect, whereas dyslexics did not. The mean amplitudes at 350-400 ms were more negative compared to those between 450 and 500 ms. The N400 effect was found between 350-500 ms, with the strongest effect between 450 and 500 ms. In addition, dyslexic and typical showed similar patterns of amplitude changes over time. Furthermore, there was a difference between the mean amplitudes in the left and middle hemisphere. Lastly, typical readers exhibited a stronger N400 response for BR compared to SF.

Name company or University: Utrecht University

UU Supervisor: Suzanne Hut

Name Student: Elisa Montana

Internship or Thesis: Thesis

Title project: Eye tracking: the Influence of Early Life Nutrition on Visual Attention

During infancy, nutrition plays an important role in developmental outcomes, including neurodevelopment and cognitive abilities. Several studies have found feeding mode to have differential effects on cognitive development and support the hypothesis that breastfeeding contributes to optimal infant neurodevelopment, compared to formula feeding. However, while many studies have used standardized neurocognitive screening tools to examine effects of infant nutrition on cognitive development, the association of early nutrition and eye-tracking technology has not been published. Visual attention, as measured by eye-tracking, is particularly valuable as an indicator for cognitive development since it provides insights into cognitive development from a much younger age, compared to other cognitive scales. The current study will assess whether the mode of feeding, being either breastfeeding or formula feeding, has an influence on infant visual attention as assessed by eye-tracking methodology. For assessment, eye-tracking data from a longitudinal study, involving over 1500 Dutch infants, will be analyzed.

Name company or University: Danone Nutricia Research

(If applicable) External supervisor: Kelly Mulder & Lidewij Schipper

UU Supervisor: Roy Hessels

Title + abstract

Workshop scientific writing

Elisa Montana

Project: thesis

Name Student: Elisa Montana

Internship or Thesis: Internship

Title project: Minimizing Cognitive Load to Enhance User Experience on Digital Platforms:
A Systematic Literature Review

This systematic literature review, conducted as part of an internship project at Daily Creations, investigated how Cognitive Load Theory can be applied to improve user experience (UX) in digital design. Focusing on the question of how minimizing cognitive load enhances user experience on digital platforms, the review analyzed studies published between 2015 and 2024. The findings highlight four key strategies for UX designers: simplifying navigation, reducing information overload, enhancing visual clarity, and improving task efficiency. By implementing these strategies, which primarily target extraneous cognitive load, designers can create more intuitive and user-friendly interfaces, leading to improved user performance, satisfaction, and overall user experience.

Name company or University: Daily Creations

(If applicable) External supervisor: Joey Verwey

UU Supervisor: Sam van Bijnen

Name Student: Koen van Oorschot

Internship or Thesis: Combined Internship/Thesis

Title project: Lights On: Designing a Fiber Photometry Specific Pre-Processing Pipeline

This thesis provides a detailed insight into the design process of a fiber photometry (FP) specific pre-processing pipeline, focusing on filtering, photobleaching correction, movement correction, and normalisation. A FIR equiripple low-pass filter was developed and tested, proving effective elimination of unwanted frequencies while maintaining signal integrity. For photobleaching correction, a double exponential decay curve fitting method was implemented, successfully eliminating trends without distorting physiological signals. Movement artefacts were mitigated using a linear regression isosbestic control signal approach, although lack of available methods prevented comparative analysis and performance assessment. Additionally, the robust Z-score transform was selected for normalisation due to its non-parametric nature, robustness against outliers, and compatibility with FP specific data dynamics. Spectral analysis, including FFT, Welch's PSD, and MODWT, validated the pipeline's effectiveness across various pre-processing stages. This report provides a transparent and reproducible framework, serving as a foundation for the FP community to further develop FP-specific pre-processing pipelines.

Name company or University: Netherlands Institute for Neuroscience

(If applicable) External supervisor: Lizz Fellingner

UU Supervisor: Sjoerd Stuit

Name Student: Martijn Boot

Internship or Thesis: Thesis

Title project: Modelling decision behaviour of cyclist in shared spaces

This thesis project investigates the decision behaviour of cyclists in shared spaces. Research has shown that cyclists in a shared space adapt their behaviour based on the density of the crowd. This could be due to visual crowding influencing their decision process. The analysis employs a drift diffusion model based on gap acceptance research to understand this behaviour in detail. Additionally, the study explores whether the density of people influences the perceived gap, and if it induces a bias in either the starting point or drift rate. This will be tested using a linear mixed effect model, with results aimed at providing deeper insights into the dynamics underlying the decision-making process.

Name company or University: Utrecht University

UU Supervisor: Floor Bontje

Name Student: Martijn Boot

Internship or Thesis: Internship

Title project: Aging, measuring cognitive aging in the aviation domain

For my internship, I examined best practices for measuring cognitive aging within the aviation domain, specifically focusing on air traffic controllers (ATCOs). I explored the potential application of the diffusion decision model as a descriptive tool for assessing the impact of aging on processing speed. To achieve this, I developed a test that simulates the everyday tasks of ATCOs and measures processing speed. Additionally, I provided advice on best practices for validating the developed test and on its potential applications. I recommended that the test be part of a three-step process, where it would first be used to identify outliers. These outliers would then be further investigated using subjective measures.

Name company or University: Royal NLR

External supervisor: Anneloes Maij (anneloes.maij@nlr.nl)

UU Supervisor: Leender van Maanen (l.vanmaanen@uu.nl)

Internship Abstract Casper Leenaars
Company: Valsplat
External Supervisor: Marloes Aben
UU Supervisor: Chris Paffen

Optimizing the 'Heroes of Value' for integration into the work of Valsplat.

This internship at Valsplat centers on the enhancement and integration of a digital-design-assessment concept called the 'Heroes of Value'—ten core values fundamental to a proper digital design, used to assess and improve digital products or services. The goal was to realize the utilization of these values within Valsplat's work and to promote their presence to clients. The project began with an exploration into the underuse of the concept, in the form of a series of interviews with colleagues. The primary research question addressed the barriers to the implementation of the Heroes of Value and how they could be effectively incorporated into the company's workflow. Due to the explorative nature of this project, the resulting methodological approach was to vary based on the outcomes of the interviews. These outcomes indicated that the underuse of the Heroes of Value was mainly due to their core values not being explained efficiently enough. This made them difficult to understand and therefore hard to use in a timely manner. The methodological approach that followed consisted of redesigning a card deck featuring the Heroes of Value for improved user experience, creating a method to embed these Heroes in research reports, and enhancing promotional strategies to sell a workshop about adding value to a digital product or service, which makes use of the Heroes of Value card deck. Preliminary results from internal pitches and customer feedback indicate a positive reception to the redesigned cards and methods. This suggests that the renewed approach could lead to greater adoption and application of the Heroes of Value in practice.

Name Student: Ting Wang

Internship or Thesis: Internship

Title project: Developing a benchmark for the Basic Approach of the EIM

Bloakes, a neuromarketing company, utilizes its Emotional Impact Model to conduct tailored market research, assessing brands/products through custom association sets and implicit association tasks. Facing inquiries about more affordable, standardized research, Bloakes introduced a "Basic Approach," with a standard association set focusing on the general population as the target audience, rather than a specific one. My internship project involved developing a benchmark and a range for the Basic Approach, and comparing it with the Custom Approach. This helps Bloakes offer clients a clearer understanding of their brand/product's emotional impact. Utilizing four diverse brands—Facebook, Ikea, Tony Chocolonely, and de Belastingdienst—to represent the upper and lower bounds of the range of the benchmark, findings revealed that the Basic Approach benchmarks significantly lower than the Custom Approach, indicating its reduced efficacy in capturing strong positive brand associations.

Name company or University: Bloakes

(If applicable) External supervisor: Rianne Besters

UU Supervisor: Famke Asberg

Name Student: Ting Wang

Internship or Thesis: Thesis

Title project: "Setting Boundaries: The Effect of Ownership and Culture on Boundary Extension"

This study investigates boundary extension, a cognitive phenomenon in which individuals recall seeing a more expansive view of a scene than was actually presented. Previous research has shown that people tend to remember scenes with wider boundaries, and this effect occurs immediately after viewing. The current study explores how boundary extension is influenced by ownership and cultural background. It examines whether owning an object affects memory of scene boundaries, hypothesizing that ownership reduces boundary extension due to increased attention to self-relevant information. Additionally, it investigates cross-cultural differences, expecting East Asians to show less boundary extension due to their holistic cognitive style, which makes them pay more attention to context. The methodology involves presenting 40 AI-generated scenes with varying zoom levels and ownership cues to 30 Chinese and 30 Western participants, assessing their memory of scene boundaries.

Name company or University: Utrecht University

UU Supervisor: Harmen Gudde

Abstract

While visual acuity testing is a highly useful tool to detect weaknesses in a person's visual system. It does not show how the observer perceives images with lower contrast (Stalin & Dalton, 2020). Patients with 20/20 visual acuity can still experience poor vision if their contrast sensitivity decreases. This is why contrast sensitivity is extremely important to be able to assess properly (Richman et al., 2013). This study looks at solving the problem of the normal contrast sensitivity measuring protocol of using 3 lengthy tests. The three tests are the low contrast ETDRS(Early Treatment Diabetic Retinopathy Study), The Vistech Vision Contrast Test System(VCTS) and the GECKO test. These tests are leading to inaccurate contrast sensitivity measurements because clients that get tested become tired of the lengthy process leading to inaccuracies in test results. Using the contrast sensitivity curve as a reference we look at the data of the 3 tests to see where on the contrast sensitivity curve they align. If we know on which areas the curve overlaps or aligns with a specific test we can in turn assign that test to a client that lands on that specific area. This would mean that one test would suffice instead of using all 3 tests. There was a significant correlation between ETDRS offsets and Gecko scores. this suggests that the Gecko test can be used to assess contrast sensitivity reliably. However, while Gecko provides valuable information, we do not have enough significant correlations in the other comparisons. This means that multiple tests are still required for a thorough assessment of contrast sensitivity on clients.

Name Student: **Deborah D.K. Duckwitz**

Internship or Thesis: **Internship**

Title project: **Unveiling Preferences in Beppy Menstrual Products Advertising – Storytelling vs. Educational Content for Potential Customers**

Abstract

This research investigates content preferences among potential customers of Beppy, a menstrual care products company based in The Netherlands, within Meta's advertising space. The study focuses on the strategic choice between educational ads and storytelling ads, acknowledging the pervasive influence of social media on brand advertising. With Meta's user base reaching 3.96 billion, and as social media continues to evolve as a pivotal channel for brand-consumer engagement, understanding preferences becomes paramount for effective marketing strategies. A survey, incorporating Likert Scale and open-ended questions, was employed. Both educational and storytelling ads were uploaded on Meta, allowing for a complete and comprehensive analysis of engagement metrics and survey results. The outcomes of the research are currently undergoing analysis.

Key words: Social Media, Menstrual Care Products, Advertising Content, Educational Content, Storytelling content, Consumer Preferences, Marketing Strategy, Brand Success, Meta.

Name company or University: **Beppy, The Netherlands**

(If applicable) External supervisor: **Sarah Cadavid and Athina Kasimidou**

UU Supervisor: **Larissa Meijer**

Name Student: Deborah Duckwitz

Internship or Thesis: Thesis

Abstract

Recent findings during the COVID-19 pandemic indicate that humans may become more prosocial in response to real-world threats. However, it is unknown whether similar effects occur with global environmental threats like climate change. This study investigates how the perception of climate change as an imminent threat influences everyday altruism and defensive emotions. We collected data from 179 participants, aged 18 to 65, assessing their self-report measures of perceived climate change threat, altruistic behavior and defensive emotions (e.g. climate change worry, and anxiety). Results indicate that higher levels of climate change worry are significantly associated with altruism. Additionally, climate change anxiety was a significant predictor of donation behavior. These results suggest that worry and anxiety triggered by climate change can motivate prosocial actions, highlighting a potential link between defensive emotional responses and altruism in the context of global environmental threats.

Keywords: Climate Change, Altruism, Prosocial Behaviour, Imminent Threat, Emotional Responses

Name company or University: Utrecht University

UU Supervisor: Joana Barbosa Vieira, Utrecht University

Name student: Isa Doeswijk

Internship or thesis: thesis

Title project: The importance of inhibitory control in everyday life is supposed by several studies, but the underlying factors influencing inhibitory control remain unclear (Kane et al., 2016; Rae, Hughes, Anderson & Rowe, 2015). The current study focuses on the neurotransmitter dopamine, which has been suggested to play an important role in the capacity of inhibitory control (Colzato et al. 2007; Gauggel et al. 2004; Mink, 1996). A stop-signal task is conducted to assess the stop-signal reaction time (SSRT), which reflects the capacity of inhibitory control (Kenemans et al., 2023). During the resting state before the performance of the stop-signal task, dopamine levels are measured by participants' eye blink rate (EBR). By combining the outcomes of the SSRT and the EBR, this study provides new insights into the relationship between dopaminergic changes and the capacity of inhibitory control.

Name company or University: Utrecht University

UU Supervisor: Sam van Bijnen

The relationship between dopamine levels and inhibitory control

Dopamine levels measured by eyeblink rates (EBRs)
Inhibitory control measured by stop-signal reaction times (SSRTs)

Inhibitory control is about the successfully inhibition of a well-practiced, automatic, or intended motor response (Curtis and D’Esposito, 2009).

Name Student: Isa Doeswijk
Internship or Thesis: Thesis cognitive inhibition
Title project: The relationship between dopamine levels and inhibitory control
Name company or University: Utrecht University
UU Supervisor: Sam van Bijnen

1 Introduction

The importance of inhibitory control in everyday life is supposed by several studies, but the underlying factors influencing inhibitory control remain unclear (Kane et al., 2016; Rae, Hughes, Anderson & Rowe, 2015). The current study focuses on the neurotransmitter dopamine which is thought to play an important role in the capacity of inhibitory control (Colzato et al. 2007; Gauggel et al. 2004; Mink, 1996). A stop-signal task is conducted to assess the stop-signal reaction time (SSRT), which reflects the capacity of inhibitory control. During the resting state before and after performance of the stop-signal task, the level of dopamine is measured by the eye blink rate (EBR) of participants (Kenemans, 2023). By combining the outcomes of the SSRT and the EBR, this study provides new insights into the relationship between dopaminergic changes and the capacity of inhibitory control.

Keywords: dopamine levels, inhibitory control, eye blink rates, stop-signal reaction time, stop-signal task

2 Methods

A popular neurocognitive task used to study inhibitory control is the so-called **stop-signal task (SST)**, introduced by Logan and Cowan (1984). In this task, subjects perform a **choice-reaction-time task** in which go-stimuli are occasionally followed by a stop-signal where the subjects have to withhold their response. The current study is based on the method presented by Kenemans and colleagues in which participants are asked to discriminate between **two go-stimuli ('X' and 'O')** by pressing left or right. As the **EBR** can be seen as a reliable and non-invasive measure of central dopaminergic functioning (Bochove et al. 2013; Bodfish et al., 1995; Colzato et al., 2009; Cruz et al., 2011; Karson, 1983; Lackner et al., 2010; Jongkees & Colzato, 2016), 2 external EEG electrodes are placed above and below the participants' eyes to measure vertical electro-oculography (vEOG). Performance on the stop-signal task (SST) is assessed using the **stop-signal reaction time (SSRT)**, which reflects the capacity of **inhibitory control**.

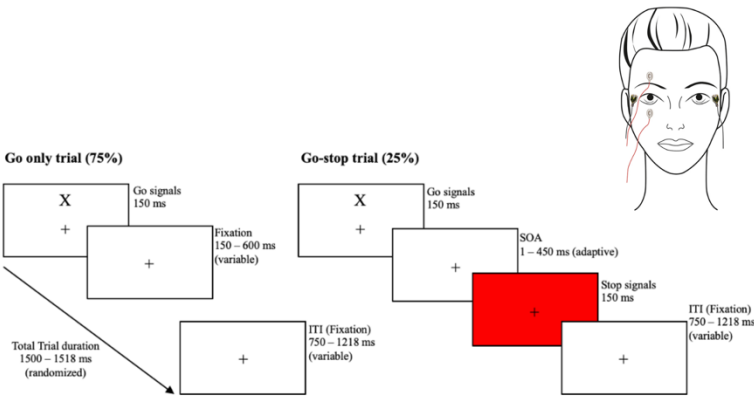


Fig. 1 Example of a go trial (left) and a stop trial (right)

3 Results

- No significant correlations were found
- Reaction times (RTs) increased from the stop-signals to the go-signals
- No relationship between the EBR and SSRT

	Mean	SD	Min	Max
SSRT (ms)	196.54	23.91	150	250
Go RT (ms)	639.45	157.36	334	1040
Stop RT (ms)	540.00	118.78	322	869
SOA (ms)	429.20	132.72	263	713
% correct	54.29	12.43	12.5	77.08
% omissions	1.49	4.75	0	31.08

Table 1 Performance results

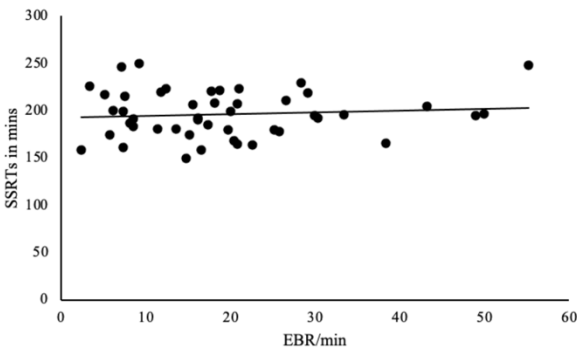


Fig. 2 Scatter diagram of individual eyeblink rate (EBR) against SSRT (in ms)

4 Conclusion

The present study investigated the relationship between dopamine levels and inhibitory control by examining the EBRs and SSRTs of 47 healthy participants between 18 and 26 years. While there was a significant variation in SSRTs of 100 ms between participants, no relationship was found between the EBR and SSRT. The current study has provided new scientific insights in that dopamine has no effect on inhibitory control in healthy people.

5 References

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Jules Convertini
Utrecht University & Netherlands Aerospace Centre
Faculty of Social and Behavioural Sciences
Program: Applied Cognitive Psychology MSc
Duration: 5 months (full-time) 27.5 ECTS
Dates: 01/02/2024 – 30/06/2024
NLR Supervisor: Maykel van Miltenburg
UU Supervisor: Dr. Chris Klink
Auditor: Dr. Martijn Mulder

ABSTRACT

This project aimed to find prosodic features in the voice that can detect and monitor cognitive workload-induced stress. Investigating prosodic features that are sensitive to stress is important, especially in aviation, for mitigating human-related errors. This study compared physiological indicators of stress to prosodic features in vocal recordings taken during an experiment where workload stress was induced. The experiment was designed to simulate stressful situations in ATC-like environments (stationary desktop environment with similar tasks to Air Traffic Controllers). At NLR, there is an ongoing project that aims to develop virtual assistants for ATC which can detect and monitor workload stress. This project will help in formulating appropriate guidelines for those virtual assistants. It was hypothesized that, during cognitive stress, there are significant correlations between self-report data, irregularities in prosodic features such as jitter, fundamental frequency, and intensity, and physiological measures of EDA, HRV, and HbO concentration in the prefrontal lateral cortex.

Name Student: Jules Convertini

Internship or Thesis: Internship

Title project: Different Tests to Assess Aspects of Contrast Sensitivity
(Abstract, 100-150 words)

Name company or University: Visio

(If applicable) External supervisor: Jan Koopman

UU Supervisor: Krista Overvliet

Contrast sensitivity is important in assessing visual impairments, and may say more about one's visual impairment than visual field or visual acuity. This paper looks at measuring contrast sensitivity via the Low Contrast ETDRS (Early Treatment Diabetic Retinopathy Study), the Vistech Vision Contrast Test System (VCTS) and the GECKO chart. The GECKO test may suffice as a measurement of contrast sensitivity, meaning patients with visual impairments might be able to complete one test, rather than multiple tests that may yield same results. This paper investigated whether the GECKO test alone would be enough to measure contrast sensitivity. Results showed that the highest points of the visual acuity charts are directly related to the contrast sensitivity assessed with the GECKO test, and that the cycles per degree as a function of contrast sensitivity is fitted by a similar line for the VCTS and ETDRS tests for the higher cycles per degree.

Tatjana Matahari Suhendra

Combined Thesis-Internship Project

“Finding the Optimal Uncertainty Communication Style and Anthropomorphism Level for Appropriate Trust in Human-Machine Teams ”

Large Language Models hold promising potential to be utilized in Human-Machine Teams within the military context. However, these models need to be designed in such a way that enables optimal collaboration, given the consequential nature of military work. Trust is an important variable in successful collaboration; manifesting behaviors such as appropriate reliance and openness to collaborate. An LLM'S ability to properly communicate uncertainty is an important aspect that influences our decision-making in regards to its suggestions, underlying appropriate reliance and establishment of long-term trust. Moreover, LLM's anthropomorphic nature of communication also influences our trust, especially in the initial stages of collaboration. Previous findings have shown that current LLM chatbots mode of uncertainty communication, their use of epistemic markers, can mislead people into over-reliance due to its less explicit nature. Additionally, the highly anthropomorphic quality of current LLM's linguistic style also breeds over-reliance, stemming from perceived-intelligence. Therefore, investigating what mode of communicating uncertainty and level of anthropomorphism fosters appropriate reliance and openness to collaboration, is incredibly pivotal to ensure the most optimal LLM design.

Company: TNO

External Supervisor: Esther Kox

UU Supervisor: Joanna Barbosa Vieira

Tatjana Matahari Suhendra
Combined Thesis-Internship Project

“Trusting Assistant Officer: Optimal LLM Chatbot Communication Style for Appropriate Trust”

Innovations in the field of AI, pertaining to LLM Chatbots have inspired conversations of implementations across industries. Given the vision of human-machine teaming in the military, these LLM Chatbots show potential for adoption. It is important that these models foster trust as it is pivotal for successful collaboration. Trust itself is calibrated through many avenues and inquiries into design features that support the calibration is necessary. This research focuses on uncertainty communication and anthropomorphic quality. The current communication style of LLM Chatbots, communicating uncertainty implicitly and being highly anthropomorphic, poses risks. Specifically for unwarranted trust. Thus, this experiment compares implicit versus explicit uncertainty communication and also high versus low anthropomorphic quality. Looking into its impact on trust alongside potential cofounders such as perceived intelligence and likeability.

Company: TNO
External Supervisor: Esther Kox
UU Supervisor: Joanna Barbosa Vieira

Name Student: Ibi Baggerman

Internship or Thesis: Thesis

Title project: **Shared Path Design: The Influence of a Cycle Lane on Cyclists' Decision-Making and Reaction Times**

This study investigates the impact of a cycle lane on cyclists' decision-making and reaction times on shared paths with different pedestrian densities. Seventeen participants performed reaction tasks within video simulations that showed cyclists through shared paths in Utrecht. Results showed that cyclists were less inclined to brake when a marked cycle path was present. However, high pedestrian density led to increased braking frequency independent of the presence of cycle path markings. Moreover, high pedestrian density significantly shortened reaction times, indicating an increased urgency to stop in crowded environments. These findings suggest that both the presence of cycle lane markings and pedestrian density are could be of influence on cyclists decision-making and reaction times, and thus relevant when designing shared paths to promote cycling and ensure safety.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Floor Bontje

Name Student: Michel Lapré
Internship or Thesis: Thesis
Title project: The ENDURE Project

For my thesis project I am currently assisting Frank Schilder in the ENDURE project, researching the effect stress has on the cognitive abilities of soldiers. My part of the ENDURE project is focused on encoding and analysing data to research the effect stress has on situational awareness and decision making specifically.

The ENDURE project consists of 68 Dutch soldiers going through two separate scenarios in the Virtual Battle Simulator 4 program while physiological measurements were measured. After both scenarios, participants were given a questionnaire to measure their situational awareness.

My part is to encode the recordings of the actions of the participants. A list of correct and incorrect behaviours was created through meetings and discussions with military personnel. Every recording was watched and behaviours will be awarded points. Any unexpected behaviours were noted and later-on potentially awarded bonus points if the behaviour was deemed correct by military personnel in a later meeting.

Name company or University: BRAIC
(If applicable) External supervisor: Frank Schilder
UU Supervisor: Gabriël Beckers

Introductie lijst:

- Endsley model of SA, DM and stress (1988).
- Positive effect of situational awareness on decision making (Stanners and French, 2003; Endsley, 1990)

- Negative effect of stress on decision making (Girotti et al., 2018). Brain!
 - Decision making worse in people with working memory struggles (Bechara, 2000)
 - Two ways of decision making (Epstein et al. 1996)
 - Bidirectional effect, gender differences (Wemm, 2017)
 - Effect of experience in stressful situations (Gamble, 2018)
 - Personality effect on negative effect of stress on decision making (Raio, 2020)
-
- Stress leads to worse focus, working memory, cognitive tunnel vision (Sneddon et al., 2013)
 - Stress leads to eepy: Bigger risks of overconfidence with bigger risks and less objective situational awareness. (Price et al.)
 - Can lead to decisions characterized by errors, greater risks and aggressive approaches.
 - And decisions made too quickly without complete information on the context and situation (Endsley et al., 2003)
-
- Early life stress can affect cognitive ability later on in life. (Pesonen et al., 2013)
 - Same, but partially reversible through social factors (Hedges and Woon, 2011).

Name Student: Yu Yang

Internship or Thesis: Thesis

Title project: The Effect of Distractor Location on (Un) Prioritized Working Memory

This thesis examined the effects of distractors and their locations on reaction time and errors in recalling prioritized versus unprioritized working memory items. Previous research generally agrees that prioritized items are recognized more quickly and accurately. However, there is debate about whether these items are protected from or vulnerable to distraction, with some suggesting that distractor location might have differing effects. In this study, 25 participants completed a working memory recall task involving two gratings and two retro-cues. Distractors appeared either at the target location, away from it, on a different side, or not at all. Results showed that reaction time and errors were lower for prioritized items. Distractors at the target location or on a different side significantly impaired performance for prioritized items, but this was not the case for unprioritized items. Thus, prioritized items are more quickly and accurately recalled but are vulnerable to distraction without a differing effect based on distractor location from unprioritized items.

Name Company or University: Utrecht University

(If applicable) External supervisor: No

UU Supervisor: Dr. Samson Chota

Name Student: Hugo (Yu Yang)

Internship or Thesis: Internship

Title project: Comparison of Online Psychological Experiment Platforms Timing

This study aims to compare the timing accuracy and precision of visual stimulus duration in different online psychological experiment platforms. Online psychological experiments are getting popular, but the platforms are not necessarily accurate and precise in timing. It is hypothesized that visual stimulus duration's timing accuracy and precision differs among online psychological experiment platforms (i.e., Gorilla, OpenSesame, Lab.js, and PsychoPy), device performances, and stimulus types. The visual stimulus durations are recorded by an oscilloscope with a photodiode, and lags and variabilities are calculated. The results show that lags differ significantly among different device performances, stimulus types, platforms, and serial positions, and variabilities differ significantly among different stimulus types and platforms. Some interaction effects on lag and variability are also discovered. In conclusion, PsychoPy has the highest accuracy and Gorilla has the highest precision. The accuracy and precision of different platforms are influenced by device performance and stimulus type. PsychoPy is the most recommended platform, and some advice on measuring the timing of experiment platforms is also given.

Name company or University: Utrecht University

UU Supervisor: Dr. Marnix Naber

Name Student: Naomi Frenkel

Internship or Thesis: Internship

Title project: Validation of the Resistance to Meat Reduction Questionnaire

The livestock sector significantly contributes to environmental damage, accounting for approximately 14.5% of global greenhouse gas emissions (Food and Agriculture Organization of the United Nations, 2017). The Resistance to Meat Reduction Questionnaire (RMRQ) measures five novel constructs regarding resistance to reducing meat intake: reactance, skepticism, inertia, social validation theory and construal level theory. Study 1 and 2 aimed to validate the scale. Study 1 revealed that men, individuals with lower educational backgrounds, and carnivores scored higher on the scale. Unexpectedly, younger individuals scored higher than older individuals. Study 2 revealed that individuals from Overijssel scored higher than individuals from Zuid-Holland. Both studies provided robust evidence for validity and reliability of the RMRQ. Study 3 will address gaps with a larger sample size, investigating the same hypotheses, the unexpected effect for *age*, potential interaction effects, high-scoring outliers on *Social Validation Theory*, and items hindering the questionnaire's/subscales' reliability.

Name company or University: De Haagse Hogeschool

External supervisor: Patricia Bulsing

UU Supervisor: Garmt Dijksterhuis

Name Student: Naomi Frenkel

Internship or Thesis: Thesis

Title project: Does a forecast's source influence our decision to act on it?: AI Algorithms versus Human Experts

Every day we navigate decisions characterised by uncertainty and complexity. We use forecasts to assist us in these decision-making processes, and these forecasts can be posed by algorithms and human experts. This paper investigates whether people are willing to take action if forecasts are posed by algorithmic sources or human expert sources, depending on three forecast characteristics (domain severity, frame and certainty level). Furthermore, it controls for human-related factors that might affect participants' choices and exploratorily investigates potential three-way interaction effects. The hypotheses assumed a tendency for algorithmic aversion based on prevailing literature. A general linear mixed model (GLMM) was used for the analyses. The results demonstrated significant interaction effects between forecast source and domain severity, as well as forecast source and certainty level. Furthermore, there was no significant interaction effect between the forecast source and frame. None of the three-way interaction effects were significant.

Name company or University: Utrecht University

UU Supervisor: Martijn Mulder

Name Student: Yi-Ching, Chen

Internship or Thesis: Internship(7.5 EC)

Title project: Relating motor imagery ability to left-right discrimination: Focus on differentiating motor imagery questionnaires and constructing left-right discrimination test using the hand-foot rotation test

Name company or University: Utrecht University

UU Supervisor: Dr. Renee Lustenhouwer

Discriminating left from right is a fundamental skill for various daily activities, yet confusion between these orientations is common among healthy individuals. Motor imagery is thought to be involved in left-right discrimination. We aim to investigate the correlation between left-right discrimination and motor imagery ability. Our primary objective is to construct an experiment which we can use it in our thesis. We using the hand and foot rotation test to discriminate left from right. It was assessed through rotation angles (0, 60, 120, 240, and 300 degrees) and viewpoints (thumb side, pinky side, palm, and back). Motor imagery questionnaires were organized into a table, detailing the tested abilities, item count, and brief descriptions. The Movement Imagery Questionnaire-3 was selected as the primary method. The outcomes of this project will provide insights into the methodologies for studying left-right discrimination and motor imagery ability.

Name Student: Yi-Ching, Chen

Internship or Thesis: Thesis

Title project: Relating motor imagery ability to left-right discrimination

Discriminating left from right is an essential skill for various daily tasks. It is a complex neuropsychologic mechanisms that engaging various cognitive processes such as visuospatial processing, memory, language, and sensory integration (Ofte & Hugdahl, 2002). Motor imagery ability is one of the factor that thought to be involved in left-right discrimination (de Lange et al., 2006, 2008). This research explores the correlation between left-right discrimination and motor imagery ability. We use hand and foot images to test the left-right discrimination, and Motor Imagery Questionnaire-3 (MIQ-3) to test the motor imagery ability. The results of our study showed a significant correlation ($r = -0.4685$, $P < 0.01$) between motor imagery and left-right discrimination.

Name company or University: Utrecht University

UU Supervisor: Dr. Renee Lustenhouwer

Name Student: Berfin Safak Sahbal

Internship or Thesis: Internship

Title project: Dynamic Multisensory Integration: Does Multisensory Input Improve Object Tracking In VR Compared To Unisensory Tracking ?

ABSTRACT

We have several senses to perceive the world, each specialized for different kinds of information. Because life's itself is dynamic, those systems integrate each other and work together during the day in order to adapt to and understand the world. However, due to the world's dynamism, it becomes hard to understand how humans deal with it by using their all senses together at the same time in an optimal level. That is why, in this project we used the Unity game engine to understand whether the multisensory input improve object tracking in VR compared to unisensory tracking. All the data was analyzed with one-way repeated ANOVA on SPSS. The results showed that visual alone stimuli is the easiest stimuli to track than audiovisual stimuli and auditory stimuli.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Nathan van der Stoep

Name Student: Frederique Graauw

Internship or Thesis: Internship

Title project: Using Design-Based Research to Develop an Exhibition Addressing the Biodiversity Crisis

Afdeling Buitengewone Zaken (ABZ) is a social design studio based in Rotterdam. ABZ specialises in design-based research addressing societal issues. This methodology combines research with design activities to create potential solutions. Naturalis, a museum dedicated to nature, requested ABZ to conduct design-based research during three days of the Dutch May holiday to answer: How can we effectively engage families with children (aged 8 to 12) in a future exhibition on the biodiversity crisis in an activating manner? After initial desk research, several frameworks were developed to explore different possible directions for this exhibition. In total, approximately 300 visitors participated in the study. One of the important findings was that visitors preferred a focus on a hopeful future and were eager to know what they could do to contribute to that. However, they did not want to be lectured. Based on all results, several recommendations were formed for Naturalis.

Name company or University: Afdeling Buitengewone Zaken

External supervisor: Nicky Liebrechts

UU Supervisor: Sam van Bijnen

Name Student: Flore Stuivenwold

Internship or Thesis: Internship

Title project: Brainbites 2: First date preferences

This research is conducted on behalf of Bloakes to create marketing content based on their own findings. For this purpose, we investigated the differences in implicit and explicit research. This is an interesting distinction to make, as implicit testing uses the unconscious part of the brain, while explicit testing uses your conscious brain. The experiment answered the question: What would you rather do on a first date?

For implicit testing, a binary choice test is performed by 200 participants, where they had to quickly choose between two images representing an activity, several times. For explicit testing, the following question is answered by 200 participants: what would you do on a first date?

Going on a walk was the primary choice when tested implicitly, while going out for dinner was the primary choice when tested explicitly. These results show us that implicit testing reveals different preferences than explicit testing.

Name company or University: Bloakes

External supervisor: Merel Schouten

UU Supervisor: Famke Asberg

Name Student: Flore Stuivenwold

Internship or Thesis: Thesis

Title project: Using a smartwatch as an external memory aid in patients with Korsakoff's syndrome is as effective as using verbal reminders for prospective memory tasks: five case studies and a meta-analysis

Korsakoff's syndrome (KS) is a chronic neuropsychiatric disorder, characterised by amnesia. KS is caused by thiamine deficiency, leading to impaired prospective memory (PM) which can lead to dangerous situations. Patients with KS therefore often reside in specialised care facilities. Within these facilities, external memory aids can be used to enhance PM functioning, increase autonomy of the patients and decrease workload for caregivers and nursing staff. A promising external memory aid is the smartwatch.

Therefore, this study investigated the effectiveness of smartwatch reminders compared to verbal reminders for task performance of PM tasks in patients with KS, and explores the patients' preferences for these reminder methods.

Five patients with KS were followed for two consecutive weeks, during which they were asked to perform four PM tasks per day, with reminders provided either by a smartwatch or verbally by the researcher. Task completion, time accuracy, episodic memory, and the patients' opinion on the used method were assessed daily. After the two experimental weeks, there is asked if the patient preferred one method over the other.

The meta-analysis showed significant results for task completion and categorical time accuracy. However, no significant differences were found for absolute time accuracy or patient preference.

Therefore, it can be concluded that smartwatch reminders are as effective as verbal reminders for task performance for patients with KS.

Name company or University: Utrecht University and Slingsdael

(If applicable) External supervisor: x

UU Supervisor: Albert Postma

Pien Carlier

Thesis:

Adaptation to misinformation

A drift diffusion analysis of bias development caused by prior information under- or overestimating probabilities.

Abstract:

In this study we explore the adaptation to under- and overestimated probabilities in prior information, focusing primarily on the development of judgmental biases, a shift in the decision criteria. In a two-alternative forced task we asked participants to indicate whether they see more black or more white in an image. Before each block, they were presented with prior information indicating varying degrees of probabilities of color dominance, either accurately reflecting the actual dominance level or under or overestimating it.

We fitted a drift diffusion model to each participants behavioral data. The parameters of the best fitting models show a larger development of a judgmental bias for strong dominance priors than for weak dominance priors, but no significant differences in bias when the prior under- and overestimates the probabilities of an event. These findings suggest that while individuals develop larger biases as the dominance in their prior knowledge strengthens, they do not effectively adjust this bias for inaccuracies in this prior.

Utrecht University, UU supervisor: Martijn Mulder

Name Student: Pien Carlier

Internship or Thesis: Internship

Title project: Towards structured qualitative experimentation: developing a clear and comprehensive format for experiment plans

Abstract:

We are developing a clear and comprehensive format for experiment plans used in Concept Development & Experimentation (CD&E), which is a structured approach to transform an idea into a viable solution ready for implementation. The format overviews all sections that should be included in the experiment plan. To encourage users of this format to thoroughly consider all sections in the format, we will include a clarification. This clarification will:

1. Clearly explain what experimentation entails.
2. Provide detailed explanations for each section, highlighting the importance of careful consideration and offering examples of the potential consequences of suboptimal choices.

We are involving all willing team members in an iterative design process. For each version of the template and the clarification, we will seek their feedback, incorporate their suggestions, and then repeat this process. This approach ensures that the final format is clear, complete, and has the full support of everyone involved.

Name company or University: Ministerie van Defensie, afdeling innovatie CLAS

(If applicable) External supervisor: Rick Douma

UU Supervisor: Leendert van Maanen